SHUBHAM DERHGAWEN

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EDUCATION

University of California, Riverside, GPA - 3.82/4 Sep 2024 – Dec 2025
Master of Science in Robotics, Focus Area: Embodied AI and Machine Learning
Courses: Advanced Robotics, Foundations of Robotics, Computational Methods for Robotics, Reinforcement Learning.

University of Pune, India, GPA - 8.43/10 Aug 2016 – Mar 2020
Bachelor of Engineering in Information Technology, First Class with Distinction
Courses: Data Structures & Algorithms, Operating Systems, Database Management Systems.

SKILLS

- **Programming Languages:** C++, Python, Bash, C#, SQL, JavaScript, Java
- **ROS:** ROS 2 Jazzy and ROS 1, C++ and Python nodes, custom messages, topics, launch files, colcon, QoS, TF2
- **DevOps & Testing:** Git, Docker, Azure DevOps CI/CD, pytest, Selenium, regression runs, Ubuntu, Agile/Scrum
- **Cloud & Data:** Azure (Data Factory, Data Lake, Batch), AWS (EC2, RDS, S3), Firebase, Node.js, Power BI
- **Autonomy:** Nav2, MoveIt, SLAM (slam_toolbox, GMapping), RRT*, A*, state machines, PyTorch, ONNX
- **Hardware & Simulation:** TurtleBot3, Arduino C++ firmware, ESP32, I2C, UART, Gazebo, Unity, CARLA

EXPERIENCE

Motion Planning Lab, UC Riverside | *Research Assistant* Sep 2025 – Present
- Developed CrowdROS, a **ROS 2** and Unity platform for testing how robots navigate around people in shared spaces.
- Wrote ROS 2 nodes in **C++** with custom message interfaces and QoS profiles between simulator and robot stack.
- Integrated unfamiliar planner codebases (DRL-VO, Arena) by running them in **Docker** to isolate dependencies.
- Built validation pipelines: unattended 75-episode **regression runs** with fixed seeds and automatic pass/fail.
- Turned failures seen in runs into repeatable, config-driven test scenarios; wrote **pytest** tests for the metric logic.
- Added recovery for unattended runs: a dropped simulator bridge or stalled episode is logged, reset, and re-run.
- Built telemetry logging and rosbag replay tools to reproduce runs and debug failures.
- Built simulated RGB camera, LiDAR, and IMU sensors and imported robots from URDF for the navigation stack.
- Trained CADRL navigation policies and ran them as ONNX models inside the ROS 2 loop.
- Built it as a tool for other developers: config-driven scenarios, no simulator code needed, documented.

Collaborative Intelligence Systems Lab, UC Riverside | *Visiting Research Student* Jul 2025 – Sep 2025
- Built a 3-vehicle, **3-machine** CARLA platform with Logitech G923 wheels for human-in-the-loop driving experiments.
- Ran a modified CARLA build in Docker that streams camera output as video instead of individual images.
- Mounted and wired vehicle sensors and operated sessions for real-world multi-agent driving data collection.

Vijeta Transport Private Limited, India | *Software Developer* Jul 2022 – Jul 2024
- Developed and maintained a **production** transport management web system with Node.js, JavaScript, and Firebase.
- Built real-time data handling and Power BI dashboards giving visibility into the **fleet**, billing, and cashflow.
- Worked directly with operators and stakeholders to turn their requirements into deployed features.
- Automated browser testing of the web system with **Selenium**.

Tata Consultancy Services, India | *Systems Engineer* Nov 2020 – Jun 2022
- Built and maintained production data pipelines in **C#** and SQL on Azure Data Factory, Data Lake, and Batch.
- Built Azure DevOps **CI/CD** pipelines with automated test and deployment stages for the pipeline code.
- Monitored, debugged, and optimized production jobs to improve reliability and reduce downtime.
- Supported migration of on-premises workflows to Azure cloud services.

PROJECTS

Autonomous Warehouse Pick-and-Deliver Robot Jan 2025
- Built a **ROS 1** pick-and-place pipeline in Gazebo with Python nodes, launch files, and a custom world.
- Sequenced navigate, pick, deliver, and place behaviors with a finite state machine coordinating base and arm.
- Mapped with GMapping (2D SLAM) and implemented RRT* from scratch for collision-free global navigation.
- Integrated MoveIt for inverse kinematics, collision-aware motion planning, and manipulator trajectory execution.

12-DOF Quadruped Robot Control Framework | *Team of 3* Jun 2025
- Wrote the Arduino **C++ firmware**: an HTTP command server driving 12 servos through a PCA9685 over I2C.
- Built a Python GUI for joint control, servo calibration, and gait testing, with link checks and request timeouts.
- Implemented inverse kinematics, a tripod gait, and a state machine that switches the robot's operating modes.
- Debugged across hardware and software with the team: traced Arduino resets under load to voltage sag.

TurtleBot3 Grid Navigation | *Foundations of Robotics, UC Riverside* May 2025
- Implemented A* grid planning in ROS, deployed over SSH to a **physical** TurtleBot3 Burger with an Intel NUC.

PUBLICATIONS

CrowdROS: A Crowd Simulation Platform for Social Robot Navigation. **Shubham Derhgawen***, Aadhar Chauhan*, Andreas Panayiotou, Julien Pettre, and Ioannis Karamouzas. *Under review, ICRA 2027.*